

Supporting Information. Nelson, M., S.C. Hosler, F.A. Boetzl, H.P. Jones, and N.A. Barber. 2020. Reintroduced grazers and prescribed fire effects on beetle assemblage structure and function in restored grasslands. *Ecological Applications*.

Appendix S1. Ground beetle traits, captures, and statistical results tables.

Table S1. Trait values for ground beetle species collected. Size is body length in mm.

species	wingtype	season	diet	activity	size
<i>Agonum muelleri</i>	macropterous	all	carnivore	diurnal	8.4
<i>Amara angustata/angustatoides</i>	macropterous	spring	herbivore	diurnal	6.6
<i>Amara crassispina</i>	macropterous	all	unknown	unknown	8
<i>Amara familiaris</i>	macropterous	all	omnivore	diurnal	6.4
<i>Amara obesa</i>	dimorphic	fall	carnivore	nocturnal	11
<i>Amara rubrica/musculis</i>	macropterous	fall	herbivore	diurnal	5.6
<i>Amara turbata</i>	macropterous	spring	unknown	unknown	9.2
<i>Anisodactylus harrisii</i>	macropterous	all	omnivore	nocturnal	11.8
<i>Anisodactylus ovularis</i>	macropterous	spring	omnivore	nocturnal	11.3
<i>Anisodactylus rusticus</i>	macropterous	spring	omnivore	nocturnal	10
<i>Bembidini</i> sp. 1	macropterous	spring	unknown	diurnal	2.5
<i>Bembidion</i> sp. 1	macropterous	spring	carnivore	diurnal	2.2
<i>Calathus gregarius</i>	dimorphic	all	carnivore	nocturnal	9.3
<i>Calathus opaculus</i>	macropterous	all	carnivore	nocturnal	9.3
<i>Chlaenius platyderus</i>	brachypterous	all	omnivore	nocturnal	14.7
<i>Chlaenius tomentosus</i>	macropterous	spring	omnivore	nocturnal	15.3
<i>Cicindela punctulata</i>	macropterous	fall	carnivore	diurnal	12.5
<i>Cicindela sexguttata</i>	macropterous	spring	carnivore	diurnal	12
<i>Cyclotrachelus seximpressus</i>	brachypterous	all	carnivore	nocturnal	16.5
<i>Cyclotrachelus sodalis</i>	brachypterous	all	carnivore	nocturnal	16.3
<i>Dicaelus elongatus</i>	brachypterous	spring	carnivore	nocturnal	17
<i>Dicheirotichus cognatus</i>	macropterous	all	omnivore	nocturnal	4.4
<i>Diplocheila obtusa</i>	macropterous	all	omnivore	nocturnal	10.7
<i>Galerita janus</i>	macropterous	all	omnivore	nocturnal	19.5
<i>Harpalus caliginosus</i>	macropterous	fall	omnivore	cathemeral	26
<i>Harpalus herbivagus/somnulentus</i>	macropterous	fall	omnivore	nocturnal	9.7
<i>Harpalus pensylvanicus</i>	macropterous	fall	omnivore	nocturnal	12.7
<i>Harpalus rubripes</i>	macropterous	all	carnivore	nocturnal	10.4
<i>Notiobia sayi</i>	macropterous	all	unknown	nocturnal	10
<i>Notiobia terminata</i>	macropterous	spring	omnivore	nocturnal	9.7
<i>Notiophilus semistriatus</i>	dimorphic	all	unknown	diurnal	5.1
<i>Ophonus puncticeps</i>	macropterous	fall	herbivore	nocturnal	7.8
<i>Poecilus chalcites</i>	macropterous	all	carnivore	nocturnal	11.8
<i>Poecilus lucublandus</i>	macropterous	all	carnivore	nocturnal	11.5
<i>Pterostichus commutabilis</i>	macropterous	all	carnivore	nocturnal	7.8
<i>Pterostichus melanarius</i>	dimorphic	all	carnivore	nocturnal	15.5
<i>Pterostichus permundus</i>	brachypterous	all	carnivore	nocturnal	15
<i>Pterostichus stygicus</i>	brachypterous	fall	carnivore	nocturnal	15.3
<i>Selenophorus ellipticus/granarius</i>	macropterous	all	unknown	nocturnal	5.9
<i>Selenophorus opalinus</i>	macropterous	all	unknown	nocturnal	10

Table S2. Results of analysis of restoration characteristics effects on community taxonomic and functional diversity metrics. Degrees of freedom = 1,14 for all F -tests.

	Activity density		Rarefied richness		Shannon diversity	
	F	P	F	P	F	P
bison	2.09	0.170	0.00	0.987	5.51	0.034
burn	2.01	0.178	0.68	0.425	0.20	0.659
age	10.96	0.005	4.60	0.050	37.83	<0.001
	adj. $R^2 = 0.507$		adj. $R^2 = 0.164$		adj. $R^2 = 0.751$	

	FRic		FEve		FDiv		FDis	
	F	P	F	P	F	P	F	P
bison	0.15	0.708	0.27	0.609	0.42	0.525	1.21	0.290
burn	0.10	0.758	0.46	0.508	0.01	0.911	0.05	0.822
age	29.17	<0.001	5.14	0.040	0.01	0.940	22.57	<0.001
	adj. $R^2 = 0.640$		adj. $R^2 = 0.162$		adj. $R^2 = 0.000$		adj. $R^2 = 0.549$	

Table S3. Results of analysis of restoration characteristics effects on individual traits in communities. Models analyzing trait proportions were binomial GLMs, so no R^2 is reported.

Degrees of freedom = 1,14 for all F -tests.

	Body size CWM		Brachypterous proportion		Carnivorous proportion	
	F	P	F	P	F	P
bison	0.18	0.679	0.31	0.589	7.21	0.018
burn	2.56	0.132	1.90	0.190	2.34	0.148
age	10.27	0.006	28.74	<0.001	7.64	0.015
adj. $R^2 = 0.417$						

	Spring proportion		Diurnal proportion	
	F	P	F	P
bison	2.09	0.170	0.21	0.657
burn	4.32	0.056	1.59	0.228
age	17.62	<0.001	8.38	0.012

Table S4. Results of distance-based redundancy analysis of ground beetle assemblages based on Bray-Curtis distances of species activity densities.

	Sum of Squares	df	<i>F</i>	<i>P</i>
bison	0.359	1	3.43	0.009
burn	0.177	1	1.70	0.126
age	0.617	1	5.90	<0.001
residuals	1.463	14		
Explained variation, constrained = 0.446, unconstrained = 0.554				

Table S5. Results of analysis of restoration characteristics effects on seed and predation rate assays. Degrees of freedom = 1,16 for all F -tests.

	July seed predation		July arthropod predation		September arthropod predation	
	F	P	F	P	F	P
bison	0.12	0.734	0.04	0.841	0.07	0.793
burn	5.12	0.038	3.46	0.082	10.95	0.004
age	1.72	0.208	0.01	0.907	0.02	0.878
	adj. $R^2 = 0.214$		adj. $R^2 = 0.025$		adj. $R^2 = 0.300$	

Table S6. Model selection results for seed predation, July arthropod predation, and September arthropod predation. R^2 is Nagelkerke's pseudo- R^2 .

Seed Predation				
Model	$\Delta AICc$	K	ω_i	R^2
Null	0.00	2	0.296	0.000
Rarefied richness	1.58	3	0.134	0.134
Proportion spring active	2.15	3	0.101	0.106
Proportion herbivore/omnivore	2.72	3	0.076	0.078
FDiv	3.21	3	0.059	0.052
FDis	3.44	3	0.053	0.040
FRic	3.86	3	0.043	0.018
Body size CWM	3.92	3	0.042	0.014
Herbivore/omnivore activity density	3.95	3	0.041	0.013
FEve	3.98	3	0.040	0.011
Total activity density	4.00	3	0.040	0.010
Proportion brachypterous	4.09	3	0.038	0.005
Proportion diurnal	4.18	3	0.037	0.000

July Arthropod Predation				
Model	$\Delta AICc$	K	ω_i	R^2
Null	0.00	2	0.329	0.000
Proportion spring active	1.65	3	0.144	0.130
Rarefied richness	2.60	3	0.090	0.083
FEve	3.64	3	0.053	0.029
Body size CWM	3.91	3	0.047	0.015
FRic	4.04	3	0.044	0.008
Total activity density	4.08	3	0.043	0.006
Proportion brachypterous	4.10	3	0.042	0.005
FDiv	4.12	3	0.042	0.004
Proportion carnivore	4.13	3	0.042	0.003
FDis	4.15	3	0.041	0.002
Proportion diurnal	4.17	3	0.041	0.001
Carnivore activity density	4.18	3	0.041	0.000

September Arthropod Predation				
Model	$\Delta AICc$	K	ω_i	R^2
Carnivore activity density	0.00	3	0.219	0.210
Null	0.20	2	0.198	0.000

Proportion carnivore	0.66	3	0.158	0.182
Total activity density	1.04	3	0.130	0.164
Proportion brachypterous	2.92	3	0.051	0.075
Body size CWM	3.04	3	0.048	0.070
Rarefied richness	3.62	3	0.036	0.040
FEve	3.91	3	0.031	0.025
FDiv	4.09	3	0.028	0.016
FDis	4.30	3	0.026	0.005
FRic	4.32	3	0.025	0.004
Proportion diurnal	4.33	3	0.025	0.003
Proportion spring active	4.37	3	0.025	0.001

Table S7. Results of analysis of restoration characteristics effects on September carnivorous ground beetle activity density. Degrees of freedom = 1,14 for all F -tests.

	F	P
bison	1.22	0.287
burn	4.98	0.043
age	4.06	0.064
adj. $R^2 = 0.396$		

Table S8. Ground beetle species capture data. Mean is average activity density per 100 trapdays, range is minimum and maximum activity densities per 100 trapdays. Bison, Fire, and Age columns summarize responses of each species to those site characteristics. Bison and Fire are percent change in activity density relative to sites without bison or prescribed fire, respectively. ‘+’ indicates species was absent in non-bison or non-fire sites, and ‘-’ indicates species were absent in sites with bison or fire. Age is correlation (Pearson’s *r*) of activity density and restoration site age in years.

Species	Total collected	Mean $\pm$ 1 s.d.	Range	Bison	Fire	Age
<i>Agonum muelleri</i>	1	0.02 $\pm$ 0.07	0–0.31	+	+	-0.29
<i>Amara angustata/angustatoides</i>	1	0.02 $\pm$ 0.07	0–0.29	–	–	-0.14
<i>Amara crassispina</i>	4	0.07 $\pm$ 0.22	0–0.9	-49	+	-0.14
<i>Amara familiaris</i>	1	0.02 $\pm$ 0.07	0–0.3	–	+	-0.11
<i>Amara obesa</i>	1	0.02 $\pm$ 0.07	0–0.29	+	–	-0.22
<i>Amara rubrica/musculis</i>	28	0.47 $\pm$ 0.58	0–2.06	147	-35	-0.48
<i>Amara turbata</i>	1	0.02 $\pm$ 0.07	0–0.29	–	+	-0.11
<i>Anisodactylus harrisii</i>	26	0.43 $\pm$ 0.56	0–1.8	-62	59	-0.20
<i>Anisodactylus ovularis</i>	2	0.03 $\pm$ 0.14	0–0.6	–	+	-0.11
<i>Anisodactylus rusticus</i>	35	0.58 $\pm$ 0.68	0–2.35	-80	-54	-0.41
<i>Bembidini</i> sp. 1	2	0.04 $\pm$ 0.11	0–0.36	90	–	0.29
<i>Bembidion</i> sp. 1	16	0.28 $\pm$ 0.69	0–2.95	284	338	0.79
<i>Calathus gregarius</i>	142	2.34 $\pm$ 2.68	0–10.29	-9	-53	-0.33
<i>Calathus opaculus</i>	176	2.9 $\pm$ 4.12	0–15	-19	-85	-0.44
<i>Chlaenius platyderus</i>	641	10.88 $\pm$ 15.08	0–57.96	457	216	0.36
<i>Chlaenius tomentosus</i>	6	0.1 $\pm$ 0.18	0–0.6	-21	1	-0.24
<i>Cicindela punctulata</i>	41	0.71 $\pm$ 0.8	0–2.14	95	81	0.51
<i>Cicindela sexguttata</i>	203	3.38 $\pm$ 2.22	0.3–7.65	11	-15	-0.31
<i>Cyclotrachelus seximpressus</i>	701	11.93 $\pm$ 12.05	0–51.48	43	108	0.80
<i>Cyclotrachelus sodalis</i>	650	11.27 $\pm$ 19.77	0.29–85.57	178	218	0.78
<i>Dicaelus elongatus</i>	13	0.21 $\pm$ 0.43	0–1.47	-71	-69	-0.20
<i>Dicheirotichus cognatus</i>	1	0.02 $\pm$ 0.07	0–0.3	+	+	0.15
<i>Diplocheila obtusa</i>	1	0.02 $\pm$ 0.07	0–0.29	–	–	-0.07
<i>Galerita janus</i>	17	0.29 $\pm$ 0.68	0–2.7	290	8	0.17
<i>Harpalus caliginosus</i>	27	0.46 $\pm$ 1.28	0–5.5	469	497	-0.22
<i>Harpalus herbivagus/somnulentus</i>	8	0.13 $\pm$ 0.27	0–0.88	160	-67	-0.21
<i>Harpalus pensylvanicus</i>	34	0.57 $\pm$ 1.93	0–8.26	659	1556	-0.28
<i>Harpalus rubripes</i>	3	0.05 $\pm$ 0.11	0–0.29	–	-50	-0.15
<i>Notiobia sayi</i>	2	0.03 $\pm$ 0.1	0–0.3	–	2	-0.18
<i>Notiobia terminata</i>	2	0.03 $\pm$ 0.14	0–0.59	–	–	-0.07
<i>Notiophilus semistriatus</i>	6	0.1 $\pm$ 0.35	0–1.47	+	400	-0.16
<i>Ophonus puncticeps</i>	46	0.76 $\pm$ 2.37	0–10.21	-93	970	-0.10
<i>Poecilus chalcites</i>	3	0.05 $\pm$ 0.11	0–0.3	-22	–	-0.15
<i>Poecilus lucublandus</i>	92	1.51 $\pm$ 1.95	0–6.76	12	263	-0.02
<i>Pterostichus commutabilis</i>	3	0.05 $\pm$ 0.15	0–0.59	–	100	0.17

<i>Pterostichus melanarius</i>	19	0.31 ± 0.39	0–1.2	-58	-28	-0.30
<i>Pterostichus permundus</i>	226	3.75 ± 2.6	0–7.21	-11	19	-0.08
<i>Pterostichus stygicus</i>	19	0.31 ± 0.44	0–1.5	-91	-9	-0.16
<i>Selenophorus ellipticus/</i> <i>granarius</i>	13	0.21 ± 0.56	0–2.35	-70	-91	0.03
<i>Selenophorus opalinus</i>	1	0.02 ± 0.07	0–0.29	–	+	-0.11

Fig. S1. Species accumulation curves for all sites.

